BID (Business Investigation Department)

Business Proposal – Accenture Innovation Challenge 2026, Round 2
Problem Track 3 — BusinessIntelligence.ai

1. EXECUTIVE SUMMARY

The modern enterprise is drowning in dashboards that show *what* happened but fail to explain *why*. When a critical metric moves, the business does not need another chart; it needs a comprehensive, evidence-backed investigation. Currently, this process relies on manual interpretation, requiring analysts to pull data across siloed systems, reconcile conflicting definitions, and hunt for anecdotal evidence before formulating a recommended action.

BID (Business Investigation Department) is a KPI intelligence-to-action engine designed to automate this investigation layer. BID transforms fragmented business data into persona-specific explanations and ranked business actions.

What sets BID apart is its **hybrid decision-intelligence architecture**. We do not treat generative AI as an oversized calculator. BID relies entirely on **deterministic analytics** to establish quantitative truth—detecting anomalies, reconciling data, and calculating driver contributions using robust statistical methods. The LLM is strictly positioned as the contextual engine: interpreting the numerical truth, grounding it in retrieved qualitative evidence, adapting the narrative for specific business personas, and phrasing actionable recommendations.

The current prototype demonstrates an end-to-end operational pipeline, complete with a React-based decision workspace, RAG-driven evidence retrieval, and persona-aware entitlement controls. By removing the manual friction from data interpretation, BID transitions organisations from descriptive Business Intelligence to true Decision Intelligence, creating a scalable blueprint for enterprise deployment.

2. BUSINESS PROBLEM

Traditional Business Intelligence tells you that a metric moved. It does not solve the operational crisis that follows.

When a critical event occurs (e.g., "Revenue falls 12%"), it triggers a manual and highly inefficient chain of events:

> **Metric moves → Analyst manually investigates → Multiple systems queried → Inconsistent definitions debated → Possible causes compared → Customer evidence checked → Business context interpreted → Action selected.**

This manual investigation loop is slow, error-prone, and suffers from several systemic failures:

- **Fragmented Data & Grains:** Analysts must reconcile daily marketing spend with order-grain transactional data and unstructured customer feedback.
- **Disconnected Dashboards:** Users stare at independent charts, struggling to correlate movements across different BI tools.
- **Lack of Evidence & Confidence:** Dashboards provide numbers without the qualitative context (like customer support tickets) necessary to validate the root cause.
- **Role-Dependent Interpretation:** A drop in conversion means something entirely different to a Marketing Manager than it does to an Operations Manager, yet both receive the exact same static report.
- **Poor Feedback Loops:** When an action is taken, the outcome is rarely fed back into the system to improve future recommendations.

A conventional dashboard stops at the anomaly. BID is designed to answer the questions that follow: *WHAT changed? WHY did it change? WHICH driver contributed most? HOW confident are we? WHAT evidence supports that? WHO needs to act? WHAT should they do? HOW should success be monitored?*

3. BID VALUE PROPOSITION

BID fundamentally changes the return on investment for enterprise data by automating the cognitive leap between noticing a problem and taking action.

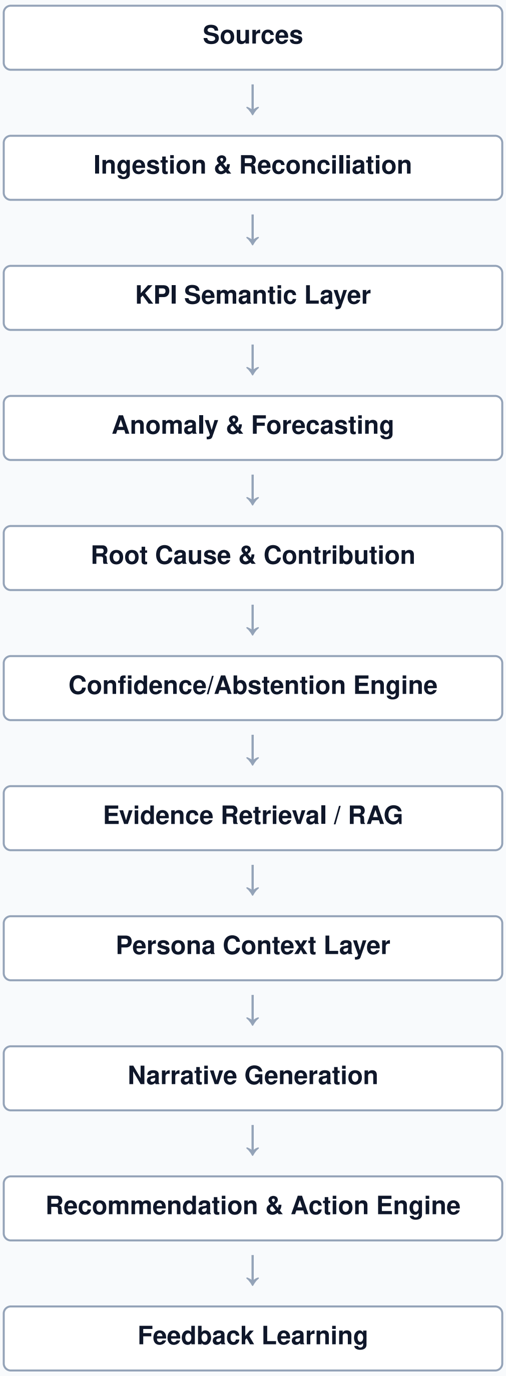
Traditional BI Pipeline:
Dashboard → Metric Movement → Manual Investigation → Manual Decision.

BID Decision Intelligence Pipeline:
Fragmented Sources → Reconciliation → KPI Intelligence → Root Cause → Evidence → Persona Context → Action → Feedback.

By automating the investigative heavy lifting, BID drastically reduces time-to-insight and analytical toil. It guarantees consistency in how metrics are interpreted, ensures that recommendations are grounded in statistical reality, and provides targeted, role-aware intelligence that empowers business leaders to pull the right levers immediately.

4. SOLUTION OVERVIEW

BID operates on a tightly governed, multi-layered architecture where quantitative processing and language generation are strictly separated.



Architecture Layers:

- **A-C. Data Ingestion & Reconciliation:** Aggregates and aligns multi-grain data across distinct sources.
- **D. KPI Semantic Layer:** Applies configuration-driven business rules, definitions, and relationships.
- **E. Anomaly Detection:** Deterministic statistical flagging (rolling z-scores, Prophet).
- **F. Root Cause Analysis:** Mathematical driver contribution and overlap calculation.
- **G. Confidence Engine:** Assesses history depth and signal strength to calibrate certainty.
- **H. Evidence Retrieval:** Date-filtered semantic search against qualitative data (e.g., reviews).
- **I-K. Narrative, Persona & Action:** LLM synthesizes the quantitative truth and evidence into role-specific narratives and ranked actions.

- **L-N. Feedback & Telemetry:** Captures analyst feedback to improve ML rankings and logs runtime metrics for observability.

5. DATA SOURCES AND HETEROGENEOUS DATA

To accurately represent business realities, intelligence cannot exist in a single-table vacuum. BID's current multi-source design ingests and processes heterogeneous data types.

The prototype demonstrates this through:

- **Order-Grain Data:** Transactional records requiring aggregation.
- **Daily Marketing Data:** Pre-aggregated spend, traffic, and campaign metrics.
- **Unstructured Evidence:** Customer reviews and support logs stored in PostgreSQL with vector embeddings.

BID uses SQL and deterministic aggregation to reconcile these different grains into a unified temporal baseline. We rely on deterministic aggregation because analytical truth must be mathematically exact. The current prototype utilizes CSV-based ingestion as a controlled boundary to demonstrate the pipeline's capabilities. The underlying reconciliation architecture is designed to seamlessly accept direct API connections, enterprise data warehouses (Snowflake, Databricks), and BI platform integrations in future enterprise expansions.

6. KPI SEMANTIC CONTRACT

A major failure point in enterprise analytics is ambiguous metric definitions. BID solves this through a configuration-driven **KPI Semantic Contract**.

Every KPI in the BID ecosystem is strictly bound to a metadata contract containing:

- KPI Name & Business Definition
- Calculation/Formula & Unit
- Drivers & Downstream Relationships
- Threshold & Direction of Good (e.g., higher revenue is good; higher CAC is bad)
- Source, Lineage, and Access Policy

This contract prevents the LLM from hallucinating calculations or misinterpreting a metric's business meaning. It ensures that regardless of who is asking or what LLM model is used, the definition, lineage, and direction of good remain immutable and globally consistent.

7. CONNECTED KPI INTELLIGENCE

A single metric rarely moves in isolation. BID satisfies the requirement for multi-KPI intelligence by mapping and analyzing connected metric trees (e.g., 3–5 connected KPIs).

Example KPI Chain:
Traffic → Visitors → Conversion Rate → Orders → AOV → Revenue
Marketing Spend → CAC → Acquisition Efficiency

Because BID understands these relationships via the Semantic Contract, it elevates the analysis. Instead of generating a shallow alert stating, *"Revenue decreased,"* BID traverses the KPI tree to deliver connected intelligence:

> *"Revenue decreased because order volume declined, while AOV remained relatively stable. Simultaneous acquisition-side changes (traffic drop and CAC spike) provide upstream context for this movement."*

This contextual bridging is a core differentiator, transforming isolated alerts into comprehensive business investigations.

8. ANOMALY DETECTION

BID relies entirely on statistical and mathematical frameworks for anomaly detection. **We do not use LLMs to detect anomalies.**

The analytical core utilizes rolling z-scores, configurable business thresholds, and Prophet forecasting to establish baselines and identify deviations. We use statistics instead of generative AI because anomaly detection demands reproducibility, auditability, and deterministic numeric truth.

In BID, *materiality* is a calculated combination of the statistical magnitude of the movement and its configurable business impact threshold. This ensures the system is sensitive to actual business pain points rather than just standard deviations, preventing alert fatigue.

9. ROOT CAUSE ANALYSIS

When an anomaly is detected, BID's quantitative engine performs automated trailing baseline comparisons and driver contribution analysis.

The system identifies which underlying drivers (e.g., product categories, geographic regions, traffic channels) mathematically overlap with the anomaly. It calculates the product-level contribution and ranks the drivers, explicitly separating primary contributors from secondary factors.

The crucial architectural distinction here is that **the LLM does not calculate root causes.** The deterministic pipeline identifies the mathematical contributors; the LLM merely translates those calculated facts into a readable narrative.

10. MULTI-FACTOR EXPLANATION

Business anomalies are rarely driven by a single factor. Consider a scenario where revenue falls. At the same time, BID detects that orders have declined, conversion has dropped, one specific product category has collapsed in sales, and customer evidence suddenly references availability issues.

BID excels at multi-factor explanation by combining the quantitative contribution analysis (the collapsed product category, the conversion drop) with supporting qualitative evidence (the customer complaints). The resulting output is a synthesized, highly credible investigation report where numerical facts and qualitative context reinforce each other seamlessly.

11. CONFIDENCE AND ABSTENTION

Confidence is not an LLM hallucination in BID; it is a computed metric. BID evaluates deterministic confidence signals, including:

- **History Depth:** Is there enough historical data to trust the baseline?
- **Credible Drivers:** Is there a clear primary driver, or are there competing, noisy signals?
- **Evidence Availability:** Does the qualitative data support or contradict the quantitative movement?

High Confidence: BID explains the root cause and recommends decisive action.
Low Confidence: BID triggers its **Abstention Engine**. It explicitly communicates uncertainty, avoids overclaiming, and requests clarification or human investigation. Abstention is a deliberate, highly valuable product feature that directly aligns with the problem statement's requirement for safe, reliable AI.

12. SPARSE HISTORY

When tracking newly launched KPIs or recently introduced products, systems often face sparse history. Conventional AI tools may attempt to extrapolate wildly or hallucinate patterns.

BID recognizes insufficient historical context mathematically. When data is sparse, the system enforces reduced confidence scores, applies cautious interpretation, explicitly states its uncertainty to the user, and refuses to fabricate certainty. This ensures the system remains a trusted advisor even when operating at the edges of available data.

13. EVIDENCE / RAG

To ground quantitative anomalies in real-world context, BID employs a Retrieval-Augmented Generation (RAG) architecture for evidence retrieval against customer reviews and support tickets.

Crucially, BID utilizes **date-first filtering** before generating embeddings and calculating cosine similarity. This matters immensely: a highly relevant complaint about "website checkout crashing" from six months ago is useless for diagnosing today's conversion drop. By enforcing temporal relevance, the evidence retrieved actively strengthens the current narrative and grounds the LLM’s output in verifiable, time-accurate reality.

14. PERSONAS

A 10% drop in revenue requires different actions depending on who is looking at the screen. BID features a persona-aware intelligence layer.

- **Marketing Manager Persona:** Receives an interpretation focused on traffic, conversion, ad spend, CAC, and channel performance.
- **Sales / Operations Manager Persona:** Receives an interpretation focused on orders, AOV, product mix, and fulfilment-adjacent issues.

BID understands that different people control different business levers. By dynamically adjusting the narrative and the recommended actions based on the user's role, BID provides actionable, role-aware decision intelligence rather than generic summaries.

15. RECOMMENDATION ENGINE

BID’s recommendation engine maps identified drivers to controllable business levers.

> **Driver → Controllable Lever → Action → Expected Impact → Owner → Confidence → Monitoring Plan**

Recommendations are drawn from a configuration-driven candidate action catalog governed by business rules. The actions are grounded entirely in actual observed drivers, purposefully avoiding generic, unhelpful AI advice (e.g., "try to increase sales"). ML ranking signals prioritize the most effective actions based on the specific operational context.

16. FEEDBACK LEARNING LOOP

Decision intelligence must improve over time. BID incorporates a closed-loop feedback mechanism.

Analysts can rate recommendations, log actions taken, and record outcomes. This feedback is stored and converted into historical scoring, which acts as an ML ranking signal to prioritize highly rated actions in future anomalies.

The current prototype demonstrates feedback-aware recommendation ranking based on the available action catalog and feedback variables. The same feedback architecture is designed to seamlessly support broader outcome learning, continuous evaluation, and richer model retraining as deployment data grows.

17. DETERMINISTIC VS LLM PROCESSING

BID's strictest architectural rule is the separation of quantitative truth from language processing.

Layer	Method	Why it is used	Why LLM is NOT used
Reconciliation	SQL / Deterministic	Ensures exact alignment of grains.	LLMs hallucinate joins and aggregations.
KPI Calculation	Math / Logic	Immutable business rules.	LLMs struggle with strict arithmetic rules.
Anomaly Detection	Statistics (Z-score, Prophet)	Auditability and reproducibility.	LLMs lack mathematical consistency.
Driver Contribution	Mathematical overlap	Exact quantification of variance.	LLMs guess numerical attribution.
Confidence	Rule-based heuristics	Safe, transparent abstention.	LLMs suffer from overconfidence.
Evidence Retrieval	Vector Search + Date Filter	Grounds insights in reality.	Prevents LLM fabrication of evidence.
Recommendation Ranking	ML / Catalog	Ensures actions tie to business levers.	Avoids generic, unactionable AI advice.
Narrative Generation	LLM	Contextualization and readability.	Pure code produces rigid, robotic text.
Action Phrasing	LLM	Persona-specific communication.	Needs linguistic adaptability for roles.

> **[NON-LLM / QUANTITATIVE TRUTH] ——** ***feeds validated data into*** **—→ [LLM / LANGUAGE + CONTEXT]**

18. SECURITY AND ENTITLEMENTS

BID utilizes a persona-aware entitlement architecture that governs identity, allowed KPIs, allowed sources, and allowed features.

The backend dynamically enforces what a user is permitted to see. For example, a Marketing Manager is not entitled to view sensitive financial margins, while Sales receives the operational information required for their decisions. The current prototype demonstrates persona-aware access patterns, with enterprise-grade field-level enforcement positioned as a seamless deployment hardening step. This enables scalable enterprise deployment without redesigning the analytics core.

19. RUNTIME TELEMETRY AND LLM ECONOMICS

AI systems must be measurable, not just accurate. BID features built-in runtime telemetry that actively tracks latency, LLM call counts, token usage, model selection, and estimated cost per insight.

This observability ensures that BID operates within realistic economic constraints. By monitoring token efficiency and tracking cost, the architecture allows for future provider substitution, caching layers, and optimization, ensuring that the system scales profitably.

20. PDF / REPORTING

BID includes a comprehensive report generation capability. When an investigation is complete, the workspace can render a formatted PDF containing the KPI movement, root cause, confidence score, evidence, narrative, recommendations, and business context.

Crucially, the PDF renderer consumes the *already-computed* analysis object. It does not needlessly trigger new LLM calls to generate the report, preserving token economics and ensuring absolute consistency between the screen and the exported document.

21. WORKSPACE / USER EXPERIENCE

BID is driven by a highly interactive React workspace designed specifically as a "decision workspace."

The UI features multiple synchronized analysis panels:

- KPI Selection and plotting
- Anomaly and root cause visualization
- RAG Evidence viewer
- Conversational assistant for deep-dives
- Custom KPI creation pane
- One-click report generation

The interface is engineered to reflect the workflow of an analyst, keeping the quantitative charts, the qualitative evidence, and the LLM narrative visible simultaneously.

22. CUSTOM KPI EXTENSIBILITY

BID allows users to define custom KPIs dynamically. Users define the KPI through its semantic contract (business name, formula, unit, threshold, direction of good, etc.).

Cross-source calculation is designed so variables from different datasets can participate in a derived KPI. It is critical to note that additional source fields act as *input variables* for calculating the derived KPI; they are not automatically treated as independent business KPIs themselves.

> **Conceptual flow:** Source Datasets → Calculation View → Derived KPI → KPI Workspace.

This provides a highly future-ready semantic extensibility framework for business users.

23. BUSINESS CASE

BID delivers measurable transformation across the intelligence lifecycle:

1. **Time-to-Insight:** Reduces multi-hour manual data hunts to near-instant automated investigations.
2. **Analyst Productivity:** Frees data teams from routine anomaly reporting to focus on strategic initiatives.
3. **Faster Operational Response:** Delivers ranked actions immediately, closing the gap between failure and correction.
4. **Role-Specific Decision Support:** Ensures the right manager gets the right levers immediately.
5. **Explainability & Consistency:** Eliminates ad-hoc, subjective interpretation of metric movements.
6. **Institutional Learning:** The feedback loop ensures that employee expertise is captured and codified into system intelligence over time.

24. TARGET MARKET

While the initial prototype is demonstrated using an e-commerce dataset because it provides a universally understood KPI chain (Traffic → Conversion → Revenue), BID is fundamentally industry-agnostic.

Because the core analytics engine is mathematical, the business semantics, personas, and action catalogs are entirely configuration-driven. This makes BID immediately applicable to SaaS (churn, MRR), retail (footfall, inventory), digital marketing (CAC, ROAS), subscription businesses, marketplaces, and general operations.

25. SCALABILITY

BID is architected for expansive growth. It is designed to scale from a single business unit to a multi-tenant enterprise deployment.

The modular architecture abstracts the ingestion and LLM provider layers, paving the way for warehouse-native operation. As deployment scales, BID is positioned to integrate directly with enterprise connectors like Snowflake, Databricks, Microsoft Fabric, and standard BI platform APIs, moving computation closer to where the enterprise data already lives.

26. LIMITATIONS AND RESPONSIBLE SCOPE

BID is built on a philosophy of responsible AI and transparent engineering. We acknowledge the following controlled prototype boundaries and their direct enterprise expansion paths:

- **Ingestion Boundaries:** The current prototype utilizes controlled CSV ingestion. *Expansion:* Production deployments will replace CSV ingestion with governed warehouse and API connectors without altering the downstream intelligence pipeline.
- **Historical Window:** The prototype operates on a limited historical dataset. *Expansion:* Cloud-native deployment will leverage full enterprise data lakes for deeper baseline generation.
- **Feedback Learning:** Recommendation learning is currently constrained by finite prototype feedback. *Expansion:* As user volume grows, the ML ranking signal will natively mature via standard reinforcement loops.

27. ROADMAP

The architecture supports a clear, phased progression:

- **Phase 1 — Intelligence Prototype (Current):** End-to-end KPI intelligence-to-action pipeline, deterministic analytics, RAG evidence, personas, recommendation ranking, feedback loops, and reporting.
- **Phase 2 — Enterprise Hardening:** Stronger entitlement enforcement, richer telemetry, broader reconciliation coverage, deployment/security hardening, API connector frameworks, and caching optimization.
- **Phase 3 — Scale:** Multi-tenant architecture, warehouse-native ingestion, enterprise connectors, broader industry configuration packs, continuous learning, and drift monitoring.
- **Phase 4 — Advanced Intelligence:** Causal inference, proactive alerting, scenario simulation, action outcome prediction, and automated monitoring execution.

28. RISK AND MITIGATION

- **Risk: Sparse History / Data Quality.** *Impact:* Erroneous baselines. *Mitigation:* The Abstention engine strictly lowers confidence and explicitly flags uncertainty.
- **Risk: LLM Cost & Rate Limits.** *Impact:* Unscalable economics. *Mitigation:* Telemetry monitoring, decoupling deterministic math from LLM calls, and future caching architectures.
- **Risk: Provider Dependency.** *Impact:* Lock-in. *Mitigation:* Agnostic prompt structuring and modular architecture allowing seamless model substitution.
- **Risk: Contradictory Evidence.** *Impact:* Confusing narratives. *Mitigation:* Quantitative truth takes precedence; semantic search enforces date-first filtering to ensure relevance.

29. DEMO SCENARIO

The Investigation Walkthrough:

1. **Detection:** BID flags a significant revenue anomaly.
2. **Magnitude & Context:** The system identifies the magnitude and inspects related KPIs (Traffic, Conversion, AOV).
3. **Root Cause & Ranking:** Deterministic driver contribution identifies and ranks the causes, highlighting a major collapse in a specific product category.
4. **Evidence Grounding:** The RAG system retrieves highly relevant, recent customer support evidence detailing a supply chain delay for that exact product.
5. **Confidence Check:** The system calculates a high confidence score due to aligning quantitative and qualitative signals.
6. **Persona Delivery (Marketing):** The Marketing Manager receives an acquisition-focused narrative advising them to pause ad spend on the out-of-stock category.
7. **Persona Delivery (Sales/Ops):** The Operations Manager receives a fulfilment-focused narrative to investigate vendor SLA breaches.
8. **Action Ranking:** The recommendation engine surfaces the highest-ranked actions for both roles.

9. **Feedback:** The analyst logs feedback on the recommendation, which the system captures to refine future ML rankings.
10. **Observability:** Runtime telemetry logs the latency, token usage, and cost of the interaction.

30. REQUIREMENT TRACEABILITY

Problem Statement Requirement	How BID Addresses It	Technology / Mechanism	Business Value
1. Detect material KPI movements	Identifies movements combining statistical deviation and business thresholds.	Rolling z-scores, Prophet forecasting.	Eliminates manual dashboard monitoring; reduces alert fatigue.
2. Reconcile context across sources	Aggregates multi-grain data (transactions, marketing, support) into unified timelines.	SQL / Deterministic aggregation.	Prevents conflicting interpretations across siloed departments.
3. Identify and rank drivers	Calculates exact variance overlap to identify primary and secondary causes.	Mathematical contribution analysis.	Removes guesswork from root-cause investigations.
4. Persona-specific narratives & evidence	Adapts explanations based on user role, grounded in date-filtered qualitative data.	LLM Narrative Engine + RAG with Cosine Similarity.	Ensures intelligence is immediately actionable and trusted.
5. Communicate uncertainty (Abstention)	Explicitly lowers confidence when data is sparse or signals conflict.	Deterministic Confidence Engine.	Builds trust; prevents AI hallucinations and unsafe decision-making.
6. Recommend practical actions	Maps drivers to controllable levers and ranks them via historical feedback.	ML-ranked Action Catalog.	Accelerates operational response time.
7. Learn from feedback	Captures user ratings and outcomes to improve future action ranking.	Closed-loop feedback storage and ML scoring.	Creates compounding institutional knowledge.
8. Realistic constraints (Security/Cost)	Implements role-based access and tracks tokens/latency per run.	Persona Entitlements & Runtime Telemetry.	Ensures enterprise scalability, security, and profitable unit economics.